

Large-Angle Statistics of the CMB Angular Correlation Function: Testing Λ CDM at Low Multipoles

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Abstract

We investigate the statistical properties of the cosmic microwave background (CMB) angular correlation function at large angular scales ($\ell \leq 30$) using Planck legacy data. Three complementary statistics are studied: the integrated squared correlation S_a^b , the interval-averaged correlation $\langle \xi \rangle_a^b$, and the antipodal correlation C_{180} . Theoretical distributions are generated via two independent methods, sampling from the MCMC posterior of the standard flat Λ CDM model fitted with the CamSpec high-multipole TT likelihood, and Monte Carlo realizations around the best-fit spectrum, allowing a direct comparison of the two approaches. We attribute the lack of large-angle correlations to a ‘look-elsewhere’ effect, as the discrepancy exists but the overall significance is reduced when considering multiple correlated statistics simultaneously.

Keywords: cosmic microwave background, angular correlation function, large-angle anomalies, Planck, Λ CDM

1. Introduction

The cosmic microwave background (CMB) provides our cleanest window onto the primordial universe. On the largest angular scales, corresponding to multipoles $\ell \lesssim 30$, the temperature anisotropy pattern carries direct information about the very largest coherent structures in the observable universe. The standard Λ CDM model predicts that the fluctuations should be a nearly Gaussian, statistically isotropic random field fully characterized by the angular power spectrum C_ℓ .

Several anomalies have been reported at low multipoles since the first-year WMAP data release (Spergel et al., 2003) and confirmed in subsequent analyses of the Planck temperature maps (Ade et al., 2013; Aghanim et al., 2020). Among the most discussed is the apparent lack of large-angle correlations: the two-point angular correlation function $C(\theta)$ is observed to be anomalously small

for $\theta \gtrsim 60^\circ$, a feature quantified by the statistic $S_{1/2}$ (Spergel et al., 2003; Copi et al., 2013). The anomaly has persisted across independent data releases and component-separation methods, yet its statistical significance remains contested. A central difficulty is that significance assessments depend sensitively on the choice of theoretical ensemble against which the data are compared, whether one fixes the cosmological parameters at their best-fit values or marginalises over the full posterior, and on whether the trial of multiple correlated statistics is properly accounted for.

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Power Spectrum & Correlation: base_CamSpecHM_TT_lowl_lowE

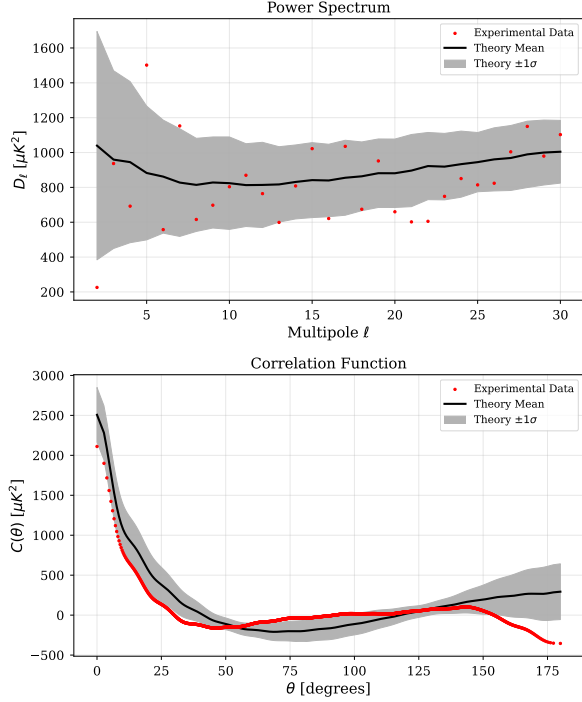


Figure 1: Power spectrum (top) and angular correlation function (bottom) for model `base_CamSpecHM_TT_lowl_lowE`. Experimental Planck data (red points), theoretical mean (black line), and $\pm 1\sigma$ band (grey) are shown. The suppression of $C(\theta)$ for $\theta \gtrsim 60^\circ$ relative to the theoretical expectation is visible in the lower panel.

Both questions have rarely been addressed simultaneously in the literature. Here we do so using Planck 2018 legacy chains: we construct the theoretical sampling distribution by two independent methods and combine the resulting p-values with the ACAT procedure (Liu and Xie, 2020), which accounts for the correlations among all statistics tested and yields a single global significance.

In this work we re-examine these statistics using Planck legacy chains and two complementary methods to construct the theoretical sampling distribution. We focus on three statistics defined over five angular intervals that partition the accessible range $[0^\circ, 180^\circ]$: the quadratic statistic S_a^b , the linear statistic $\langle \xi \rangle_a^b$, and the antipodal correlation C_{180} .

The primary model considered throughout is the standard flat Λ CDM fit to the Planck temperature power spectrum using the CamSpec high-multipole TT likelihood combined with the low- ℓ temperature and polarization likelihoods (Aghanim et al., 2020).

Results for the Plik-based likelihood and for models with free curvature Ω_k are reported in the appendix.

2. Statistics

The statistics studied here are designed to probe complementary aspects of the large-angle correlation function: its integrated power, its mean level over specific angular ranges, and its value at antipodal separation. The choice is guided by the existing literature on low- ℓ anomalies, though as noted by Efstathiou et al. (2010), some statistics, including $S_{1/2}$, were identified a posteriori from the data, which inflates their apparent significance. Studying multiple statistics simultaneously, and combining their p-values through a global test, is therefore essential to obtain a reliable assessment of tension with Λ CDM.

2.1. Angular Power Spectrum and Correlation Function

The CMB temperature anisotropy $\Delta T(\hat{n})/T_0$ is expanded in spherical harmonics with coefficients $a_{\ell m}$. For a statistically isotropic Gaussian field the two-point statistics are fully encoded in the angular power spectrum $C_\ell = \langle |a_{\ell m}|^2 \rangle$. We work with the conventional scaled spectrum $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$, which is approximately flat for a scale-invariant primordial spectrum.

The angular two-point correlation function is obtained by transforming back to real space via a Legendre expansion,

$$C(\theta) = \sum_{\ell} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos \theta) = \sum_{\ell} \frac{2\ell+1}{2\ell(\ell+1)} D_\ell P_\ell(\cos \theta), \quad (1)$$

where P_ℓ are Legendre polynomials. The sum runs from $\ell = 2$ to $\ell_{\text{max}} = 30$ throughout this work.

2.2. Interval Statistics

Three statistics are evaluated on each of the angular intervals listed in Table~1, chosen to partition the accessible range $[0^\circ, 180^\circ]$ into physically motivated regions while including the classic $S_{1/2}$ interval for comparison with the literature.

Antipodal correlation $C_{180} \equiv C(\theta = 180^\circ)$ measures the correlation between antipodal points on the sky. Because it is a pointwise evaluation of

$C(\theta)$ it is relatively insensitive to masking effects at intermediate scales, and it provides a direct test of the near-vanishing of large-angle correlations at the largest possible separation.

Interval-averaged correlation $\langle \xi \rangle_a^b$ is the mean of $C(\theta)$ over the angular interval $[\theta_a, \theta_b]$, expressed in terms of $\mu = \cos \theta$ as

$$\langle \xi \rangle_a^b = \frac{1}{b-a} \int_a^b C(\theta) d\mu = \sum_{\ell} \frac{D_{\ell}}{2\ell(\ell+1)} \frac{P_{\ell+1}(b) - P_{\ell-1}(b) - P_{\ell+1}(a) + P_{\ell-1}(a)}{b-a}, \quad (2)$$

where $a = \cos \theta_b$ and $b = \cos \theta_a$. Being linear in D_{ℓ} , this statistic is sensitive to shifts in the mean correlation level within an interval and changes sign when the correlation function crosses zero, making it complementary to S_a^b .

Integrated squared correlation S_a^b is defined as

$$S_a^b = \int_a^b C^2(\theta) \sin \theta d\theta = \sum_{n,m} A_n A_m D_n D_m T_{nm}, \quad (3)$$

where $A_{\ell} = (2\ell+1)/[2\ell(\ell+1)]$ and

$$T_{nm} = \sum_{r=0}^{\min(m,n)} \frac{A_r A_{m-r} A_{n-r} [\Delta P_{m+n-2r+1} - \Delta P_{m+n-2r-1}]_a^b}{A_{m+n-r} (2m+2n-2r+1)}$$

are overlap integrals of Legendre polynomials over the interval (Carlitz, 1961). Being quadratic in D_{ℓ} , this statistic amplifies deviations that are coherent across multipoles and is always non-negative, so it cannot be cancelled by sign changes of $C(\theta)$ within the interval. The classic $S_{1/2}$ statistic of Spergel et al. (2003) corresponds to the choice $[\theta_a, \theta_b] = [60^\circ, 180^\circ]$.

3. Data

Table 1: Angular intervals used for S_a^b and $\langle \xi \rangle_a^b$.

Angular range	Description	$[\cos \theta_b, \cos \theta_a]$
$[0^\circ, 30^\circ]$	Small angles	$[0.865, 1.000]$
$[30^\circ, 92^\circ]$	Intermediate	$[-0.039, 0.865]$
$[92^\circ, 154^\circ]$	Large angles	$[-0.901, -0.039]$
$[154^\circ, 180^\circ]$	Near-antipodal	$[-1.000, -0.901]$
$[60^\circ, 180^\circ]$	Classic $S_{1/2}$	$[-1.000, 0.500]$

3.1. Planck Posterior Chains

We use the publicly available Planck 2018 MCMC chains from the Planck Legacy Archive¹ (Planck Collaboration, 2015). The primary dataset is the CamSpec high-multipole TT likelihood combined with the low- ℓ temperature and polarization likelihoods (base_CamSpecHM_TT_lowl_lowE). This provides posterior samples of the six standard Λ CDM parameters $\{\Omega_b h^2, \Omega_c h^2, H_0, \tau, n_s, A_s\}$ along with a suite of derived quantities. The Plik-based TT likelihood and two curvature-extended runs are considered in Appendix~Appendix A.

For each posterior sample the theoretical power spectrum D_{ℓ} is computed with CAMB (Lewis), and the corresponding $C(\theta)$ and derived statistics are evaluated analytically using equations 1–3.

3.2. Observed Planck Multipoles

The experimental values of each statistic are computed from the Planck 2018 Commander temperature map and associated beam and pixel window functions, using the same multipole range $\ell \in [2, 30]$. These observed values serve as the reference against which the theoretical distributions are compared.

3.3. Ensemble Construction

Two independent strategies are used to construct the theoretical sampling distribution.²

1. **MCMC ensemble.** For each of the 1000 samples drawn from the tail of the Planck posterior chain, specifically the last 1000 rows after GetDist’s default burn-in removal, the cosmological parameters are passed to CAMB to compute a theoretical spectrum D_{ℓ}^{th} . Each spectrum is then passed through the simulation pipeline described below, producing one

¹<https://pla.esac.esa.int/>

²The full analysis pipeline is publicly available at <https://github.com/PhyAMR/CMB>.

realisation of cosmic variance on top of the parameter-determined D_ℓ^{th} . The resulting ensemble therefore captures both parameter uncertainty and cosmic variance simultaneously, constituting a Bayesian predictive distribution for each statistic.

2. **Best-fit ensemble.** The best-fit D_ℓ^{bf} is read from the `.minimum.theory.cl` file of the Planck chain, which contains the spectrum evaluated at the maximum-posterior parameter point. This single spectrum is used as input to 1000 independent pipeline realisations, each with a different random seed. The spread of the resulting ensemble is driven entirely by cosmic variance at fixed cosmological parameters, with no contribution from parameter uncertainty. The width of its histograms is therefore a lower bound on the total variance: the additional width seen in the MCMC histograms is attributable purely to parameter uncertainty, and comparing the two ensembles provides a clean decomposition of the two contributions.

In both cases, the pipeline follows the methodology of [Sullivan et al. \(2024\)](#). For each theoretical C_ℓ^{th} spectrum, the beam transfer function b_ℓ and the HEALPix pixel window function p_ℓ are applied, a full-sky temperature map is synthesised at $N_{\text{side}} = 2048$ using `hp.synfast`, and the pseudo- C_ℓ are recovered with `hp.anafast`. The recovered spectrum is then deconvolved by $b_\ell^2 p_\ell^2$ to obtain an estimate $\hat{D}_\ell$ in the same space as the published Planck values. When a sky mask is applied, the recovered C_ℓ are corrected by the sky fraction f_{sky} before deconvolution. Simulations were run both with the Planck 2018 common mask and without any mask; results for both cases are reported. At $\ell_{\text{max}} = 30$ the beam and pixel window corrections are numerically negligible, but they are included for consistency with the Planck data reduction.

In order to reduce the look-elsewhere effect and obtain a calibrated global significance that accounts for the correlation structure of the statistics without requiring an explicit model of it, we use the Aggregated Cauchy Association Test (ACAT; [Liu and Xie, 2020](#)). The eleven statistics examined, five intervals each for S_a^b and $\langle \xi \rangle_a^b$, plus C_{180} , are not independent: all derive from the same realisation of D_ℓ , so their p-values are correlated by construction. Treating them as independent and applying a simple Bonferroni correction would be overly con-

servative, while reporting only the most extreme individual p-value would be anti-conservative.

For each statistic $k \in \{1, \dots, 11\}$ the individual p-value p_k is computed as the two-sided empirical quantile of the observed value within the simulated ensemble: $p_k = 2 \min(F_k, 1 - F_k)$, where F_k is the fraction of realisations with a statistic value less than or equal to the observed one. The ACAT test statistic is then

$$T = \sum_{k=1}^{11} w_k \tan\left[\left(\frac{1}{2} - p_k\right) \pi\right],$$

where equal weights $w_k = 1/11$ are used, reflecting no a priori preference for any particular angular interval or statistic. Under the null hypothesis, T follows a standard Cauchy distribution regardless of the dependence structure among the p_k , so the combined p-value is obtained analytically as

$$p_{\text{global}} = \frac{1}{2} - \frac{\arctan T}{\pi}.$$

The Cauchy combination is exact under arbitrary dependence, including the strong correlations expected between statistics computed over overlapping angular ranges. The global tension in units of Gaussian standard deviations is reported as $n_\sigma = \Phi^{-1}(1 - p_{\text{global}}/2)$, where Φ^{-1} is the probit function.

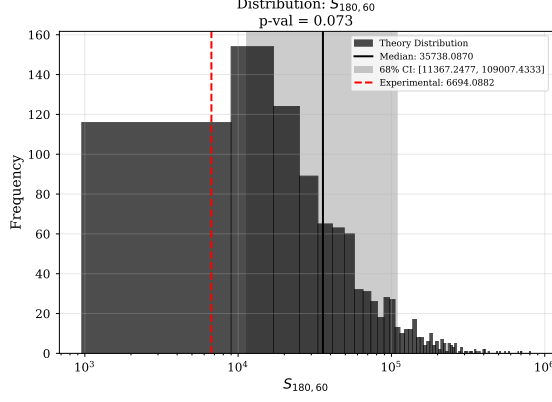
4. Results

4.1. Best-Fit Analysis

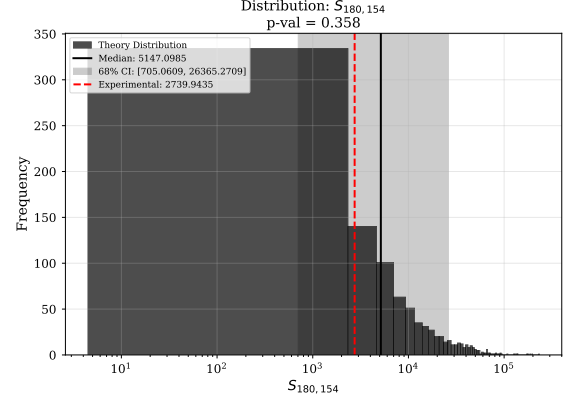
In the best-fit analysis the theoretical sampling distribution is generated by Monte Carlo realizations of the CMB sky drawn from the best-fit Λ CDM spectrum. This approach isolates cosmic variance from parameter uncertainty.

Figures 2 and 3 display the distributions of the S_a^b and $\langle \xi \rangle_a^b$ statistics respectively, for all five angular intervals. Each panel shows the distribution over the 1000 Monte Carlo realizations (blue histogram), and the observed Planck value (red vertical line). Figure 4 shows the analogous distribution for C_{180} .

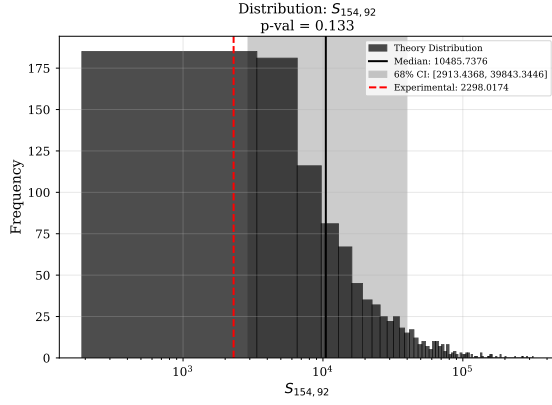
As we can see, for the different intervals all values of S_a^b are below the median of the distribution, which is not only consistent with the lack of large-angle correlations but also suggests that the lack is



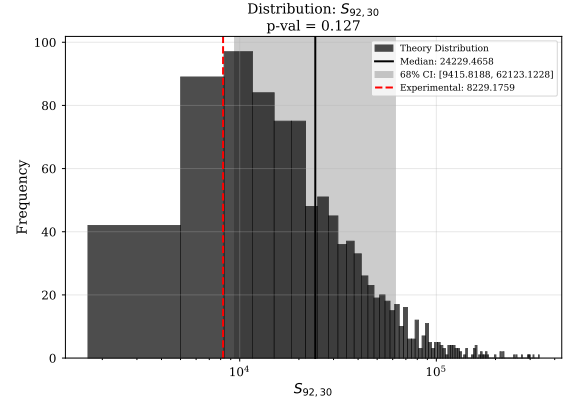
(a) S_{60}^{180} (classic $S_{1/2}$)



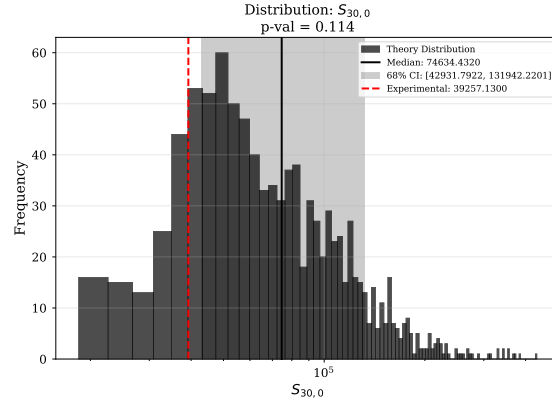
(b) S_{154}^{180}



(c) S_{92}^{154}



(d) S_{30}^{92}



(e) S_0^{30}

Figure 2: Best-fit Monte Carlo distributions of the S_a^b statistic for model **base_CamSpecHM_TT_low1_lowE** ($\ell_{\max} = 30$). Each panel shows the normalized histogram of 1000 realizations; the red line marks the observed Planck value. The x-axis is in logarithmic scale.

not only present at large angles but also at intermediate and small angles. Nevertheless, the observed values are not in the extreme tails of the distributions, and the empirical p-values are not low enough to claim a significant tension with Λ CDM.

In this set of histograms we can see that the observed values of $\langle \xi \rangle_a^b$ sometimes are lower and sometimes higher than the median of the distribution, which is consistent with the fact that $\langle \xi \rangle_a^b$ is a linear statistic, and therefore it is more sensitive to shifts in the mean correlation level, and the change of sign in the correlation function in certain intervals.

The statistical summary , including the median, 68% credible interval, observed value, and empirical p-value for each statistic , is reported in Table~2.

4.2. MCMC Analysis

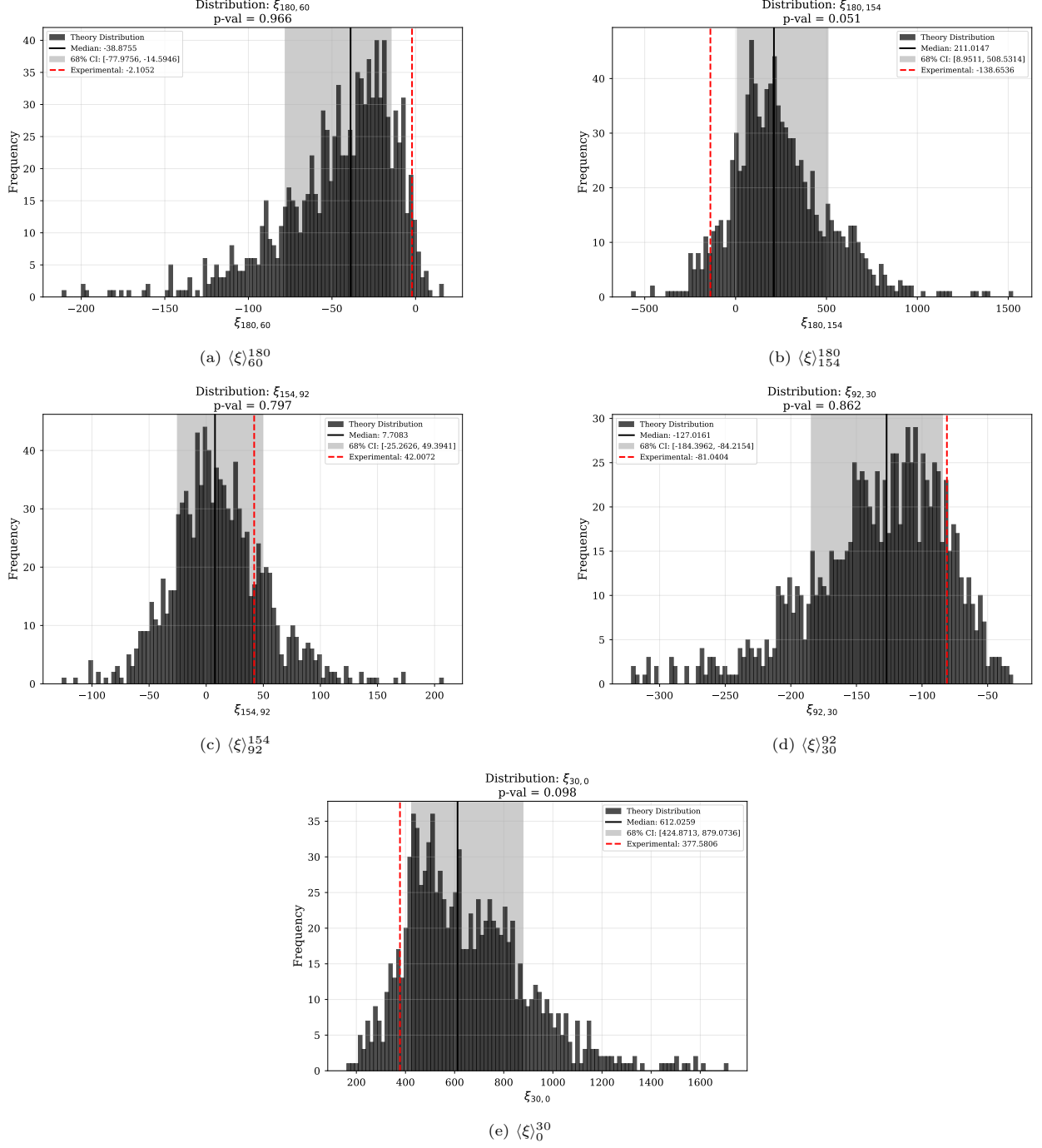


Figure 3: Best-fit Monte Carlo distributions of the interval-averaged correlation $\langle \xi \rangle_a^b$ for model **base_CamSpecHM_TT_low1_lowE**.

Table 2: Statistical summary for the best-fit analysis of model `base_CamSpecHM_TT_lowl_lowE`. Columns give the experimental value, best-fit median, 68% interval, and empirical p-value.

Global tension (ACAT): $T = 1.966$, $p = 1.50e - 01$, 1.44σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	260.7976	$[-38.2572, 629.7045]$	0.022
$S_{180,60}$	6694.0882 ± 8822.2915	35725.0265	$[11308.2691, 108785.4640]$	0.073
$S_{180,154}$	2739.9435 ± 3018.8131	5144.2773	$[704.4897, 26362.7403]$	0.358
$S_{154,92}$	2298.0174 ± 3103.8229	10463.2839	$[2893.9033, 39807.8958]$	0.133
$S_{92,30}$	8229.1759 ± 10643.1270	24227.7116	$[9406.5535, 62090.4792]$	0.127
$S_{30,0}$	$39257.1300 \pm 48919.8333$	74487.4419	$[42884.9866, 131879.2429]$	0.114
$\xi_{180,60}$	-2.1052 ± 12.1260	-38.8758	$[-78.4081, -14.6241]$	0.966
$\xi_{180,154}$	-138.6536 ± 57.5605	210.8664	$[7.7966, 508.4919]$	0.051
$\xi_{154,92}$	42.0072 ± 34.6449	7.6378	$[-25.2907, 49.3801]$	0.797
$\xi_{92,30}$	-81.0404 ± 79.0240	-127.2322	$[-184.4908, -84.2408]$	0.862
$\xi_{30,0}$	377.5806 ± 351.5645	611.8525	$[424.7966, 878.9550]$	0.098

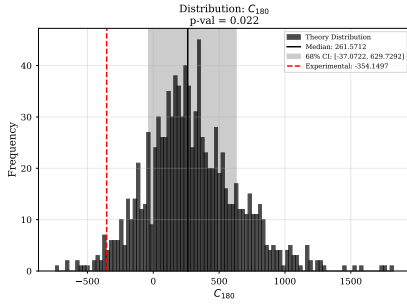


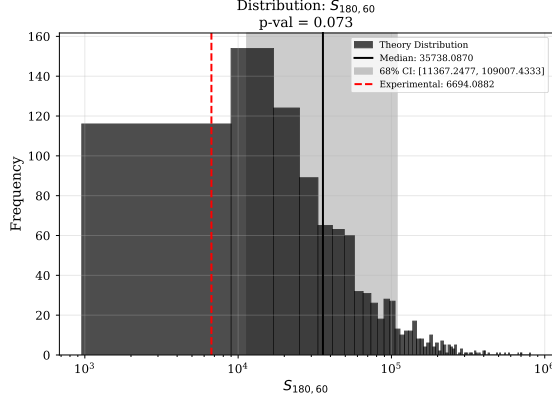
Figure 4: Best-fit Monte Carlo distribution of the antipodal correlation C_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

In the MCMC analysis the theoretical sampling distribution is constructed directly from the Planck posterior, propagating full parameter uncertainty through to the derived statistics. The 1000 posterior samples drawn from the CamSpec Λ CDM chain each yield a distinct D_ℓ spectrum, whose distribution reflects both cosmic variance and parameter degeneracies.

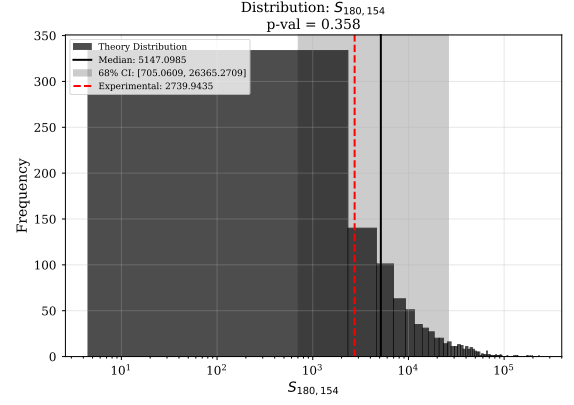
Figures 5, 6, and 7 show the resulting distributions of S_a^b , $\langle \xi \rangle_a^b$, and C_{180} respectively. The broader width of the MCMC-derived histograms compared to the best-fit case reflects the additional spread due to cosmological parameter uncertainty.

The quantitative results for the MCMC ensemble are collected in Table 3.

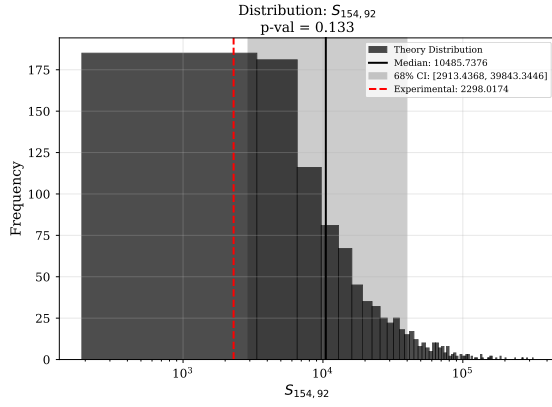
5. Discussion



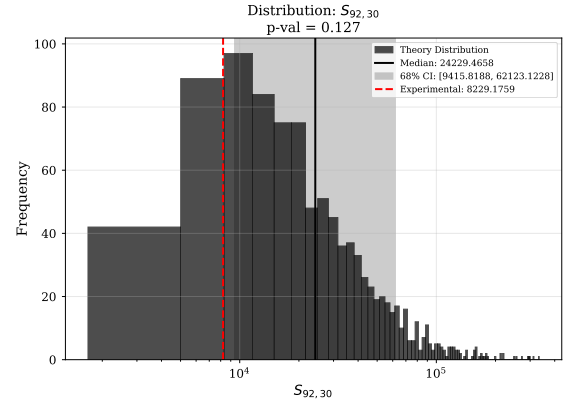
(a) S_{60}^{180} (classic $S_{1/2}$)



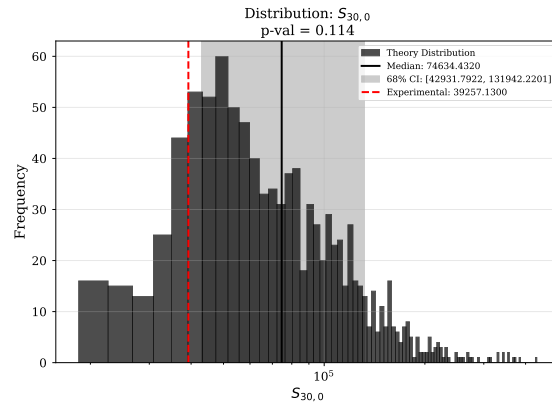
(b) S_{154}^{180}



(c) S_{92}^{154}



(d) S_{30}^{92}



(e) S_0^{30}

Figure 5: MCMC posterior distributions of the S_a^b statistic for model **base_CamSpecHM_TT_low1_lowE**. Each panel shows the distribution over 1000 posterior samples from the Planck CamSpec chain.

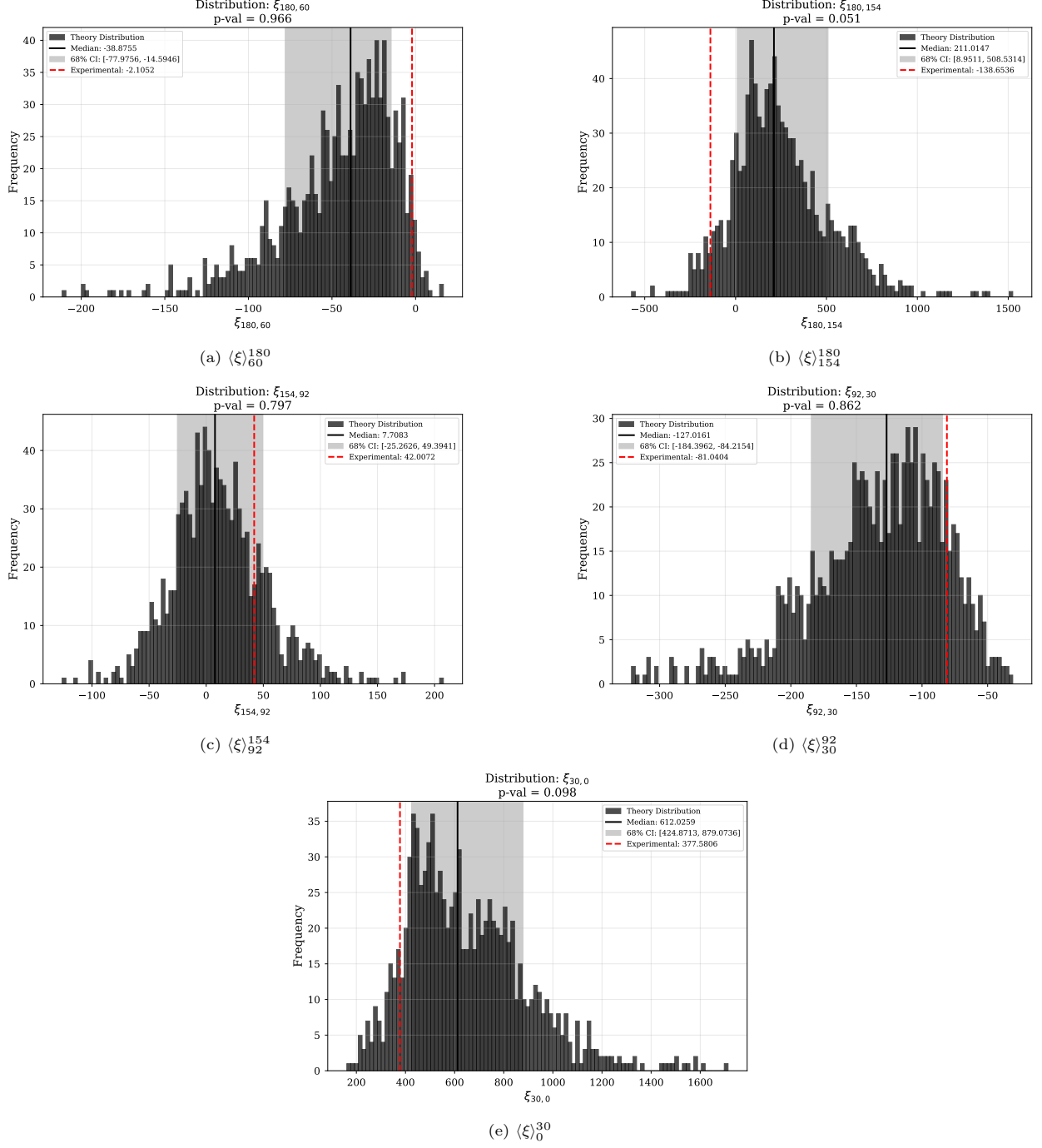


Figure 6: MCMC posterior distributions of $\langle \xi \rangle_a^b$ for model **base_CamSpecHM_TT_low1_lowE**.

Table 3: Statistical summary for the MCMC analysis of model `base_CamSpecHM_TT_low1_lowE`. Columns give the experimental value, MCMC median, 68% credible interval, and empirical p-value.

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Statistic	Experimental	MCMC Median	68% CI	p-value
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$S_{180,60}$	6694.0882 ± 8822.2915	33937.6161	$[10094.0365, 102450.9702]$	0.076
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$S_{154,92}$	2298.0174 ± 3103.8229	10088.0224	$[2572.3400, 34735.6347]$	0.138
$S_{92,30}$	8229.1759 ± 10643.1270	23088.7680	$[9244.2759, 60281.6008]$	0.137
$S_{30,0}$	$39257.1300 \pm 48919.8333$	72409.8155	$[41887.8294, 128755.4170]$	0.133
$\xi_{180,60}$	-2.1052 ± 12.1260	-37.1448	$[-77.0816, -13.6870]$	0.962
$\xi_{180,154}$	-138.6536 ± 57.5605	210.0608	$[-7.9407, 473.6089]$	0.047
$\xi_{154,92}$	42.0072 ± 34.6449	8.7805	$[-22.1704, 48.5205]$	0.802
$\xi_{92,30}$	-81.0404 ± 79.0240	-126.3688	$[-182.5317, -81.9472]$	0.844
$\xi_{30,0}$	377.5806 ± 351.5645	603.5009	$[414.7549, 861.6303]$	0.116

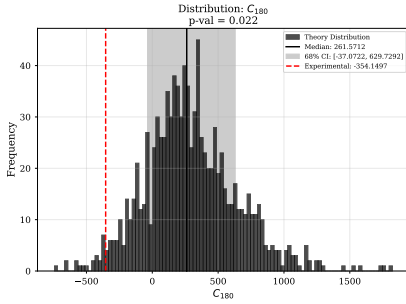


Figure 7: MCMC posterior distribution of C_{180} for model `base_CamSpecHM_TT_low1_lowE`.

The results for alternative likelihoods and models with free curvature Ω_k are reported in Appendix~[Appendix A](#). The dependence of the results on the likelihood choice (CamSpec vs. Plik) provides a useful cross-check, while the curvature-extended models test whether a departure from flatness could alleviate the observed tensions.

As noted in ([Efstathiou et al., 2010](#)), in the low- ℓ regime, the theoretical distribution of the statistics is mostly independent of the mask used, and the effects of the mask are negligible.

6. Conclusions

We have studied the large-angle CMB angular correlation function and its derived statistics using Planck 2018 data and two complementary ensembles for the theoretical sampling distributions. The statistics examined, the quadratic S_a^b , the linear

$\langle \xi \rangle_a^b$, and the antipodal C_{180} — probe distinct aspects of the large-angle temperature field.

Both ensembles provide complementary perspectives on the significance of the large-angle anomalies. The best-fit ensemble isolates the contribution of cosmic variance, while the MCMC ensemble captures the full parameter uncertainty and the cosmic variance.

A complete summary of all p-values and multivariate tension estimates for the primary `base_CamSpecHM_TT_low1_lowE` model is provided in Tables~[2](#) and [3](#); results for additional models appear in Appendix~[Appendix A](#).

While the lack of large-angle correlations is a persistent feature of the observed CMB sky, specially the C_{180} statistic, the Λ CDM model is not in significant tension with the data when the full distribution of statistics is taken into account. The observed values are generally below the median of the theoretical distributions, consistent with the lack of large-angle correlations.

Another raised question from previous analyses is whether the lack of large-angle correlations is a statistical fluke or a hint of new physics. The observed p-values for the C_{180} statistic are low accross multiple simulations and models, and while they are in the order of 2-2.5 σ the fact that it is persistent accross all our simulations makes it more clear that the lack of large-angle correlations is a real feature of the data, and not a statistical fluke.

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Appendix A. Additional Models: Statistical Tables

The tables in this appendix report the full statistical summary for three additional Planck models. All quantities are defined as in the main text. Model names follow the Planck convention: `omegak` denotes a run with free spatial curvature; `plikHM` denotes the Plik high-multipole likelihood; `TTTEEE` denotes a joint temperature and polarization fit.

Appendix A.1. No masked sky

In this section the theoretical sampling distributions are constructed without applying any sky mask, so the recovered C_ℓ are not corrected by f_{sky} before deconvolution. The observed values are computed from the full-sky Commander map, so they are directly comparable to the theoretical distributions without any correction for masking effects.

Appendix A.1.1. MCMC Results

Table A.4: MCMC results for `base_plikHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 1.750$, $p = 1.65e - 01$, 1.39σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	241.2170	$[-55.1699, 594.4383]$	0.029
$S_{180,60}$	6694.0882 ± 8822.2915	33801.8449	$[11110.7265, 102182.3499]$	0.065
$S_{180,154}$	2739.9435 ± 3018.8131	4766.0801	$[672.8630, 22097.9814]$	0.376
$S_{154,92}$	2298.0174 ± 3103.8229	10527.3578	$[2848.7506, 34955.3646]$	0.125
$S_{92,30}$	8229.1759 ± 10643.1270	23310.8042	$[9398.8164, 59071.8598]$	0.128
$S_{30,0}$	$39257.1300 \pm 48919.8333$	73229.9679	$[42537.9015, 125384.8849]$	0.126
$\xi_{180,60}$	-2.1052 ± 12.1260	-38.5341	$[-76.0073, -14.4676]$	0.962
$\xi_{180,154}$	-138.6536 ± 57.5605	202.3885	$[-6.9888, 468.8826]$	0.052
$\xi_{154,92}$	42.0072 ± 34.6449	9.1633	$[-25.4628, 51.1022]$	0.781
$\xi_{92,30}$	-81.0404 ± 79.0240	-126.3941	$[-178.6658, -84.8224]$	0.860
$\xi_{30,0}$	377.5806 ± 351.5645	607.6992	$[416.0714, 851.9444]$	0.092

Table A.5: MCMC results for `base_omegak_CamSpecHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 1.841$, $p = 1.58e - 01$, 1.41σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	208.9942	$[-34.4807, 486.2964]$	0.011
$S_{180,60}$	6694.0882 ± 8822.2915	21765.8299	$[8109.5135, 74119.3517]$	0.119
$S_{180,154}$	2739.9435 ± 3018.8131	3267.8897	$[391.6000, 15964.8461]$	0.466
$S_{154,92}$	2298.0174 ± 3103.8229	7021.0369	$[2079.4052, 26823.1748]$	0.186
$S_{92,30}$	8229.1759 ± 10643.1270	15837.8991	$[6161.9693, 41989.8915]$	0.246
$S_{30,0}$	$39257.1300 \pm 48919.8333$	51438.9870	$[29716.1148, 94216.3869]$	0.310
$\xi_{180,60}$	-2.1052 ± 12.1260	-30.7119	$[-64.0166, -12.3069]$	0.957
$\xi_{180,154}$	-138.6536 ± 57.5605	164.2943	$[-0.5697, 396.1609]$	0.030
$\xi_{154,92}$	42.0072 ± 34.6449	6.7193	$[-20.7478, 41.0697]$	0.847
$\xi_{92,30}$	-81.0404 ± 79.0240	-105.5266	$[-152.8439, -67.5677]$	0.734
$\xi_{30,0}$	377.5806 ± 351.5645	502.4040	$[340.1959, 725.3792]$	0.232

Table A.6: MCMC results for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

Global tension (ACAT): $T = 1.728$, $p = 1.67e - 01$, 1.38σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	208.6258	$[-45.5817, 506.9834]$	0.013
$S_{180,60}$	6694.0882 ± 8822.2915	25836.5779	$[8014.3153, 74173.0515]$	0.111
$S_{180,154}$	2739.9435 ± 3018.8131	3617.6754	$[464.5482, 16839.1825]$	0.455
$S_{154,92}$	2298.0174 ± 3103.8229	7418.2542	$[2223.0737, 24954.1641]$	0.172
$S_{92,30}$	8229.1759 ± 10643.1270	18231.4590	$[6845.5271, 42367.6853]$	0.205
$S_{30,0}$	$39257.1300 \pm 48919.8333$	55436.6264	$[32260.9180, 96229.5173]$	0.264
$\xi_{180,60}$	-2.1052 ± 12.1260	-33.4438	$[-63.2413, -11.6939]$	0.961
$\xi_{180,154}$	-138.6536 ± 57.5605	168.6679	$[-15.5933, 403.9776]$	0.040
$\xi_{154,92}$	42.0072 ± 34.6449	9.8408	$[-20.0036, 45.5296]$	0.816
$\xi_{92,30}$	-81.0404 ± 79.0240	-110.8947	$[-155.6151, -70.9814]$	0.766
$\xi_{30,0}$	377.5806 ± 351.5645	522.1450	$[353.3445, 736.3491]$	0.196

Appendix A.1.2. Best-Fit Results

Table A.7: Best-fit results for `base_plikHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 1.554$, $p = 1.82e - 01$, 1.33σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	235.3880	$[-62.2566, 586.2191]$	0.028
$S_{180,60}$	6694.0882 ± 8822.2915	33261.4972	$[10608.9403, 100001.9703]$	0.082
$S_{180,154}$	2739.9435 ± 3018.8131	4638.8774	$[579.8045, 21964.4118]$	0.379
$S_{154,92}$	2298.0174 ± 3103.8229	9746.7383	$[2727.4077, 35795.7064]$	0.130
$S_{92,30}$	8229.1759 ± 10643.1270	23100.4791	$[9000.3718, 60298.5831]$	0.145
$S_{30,0}$	$39257.1300 \pm 48919.8333$	70442.7090	$[40918.0369, 130962.8091]$	0.137
$\xi_{180,60}$	-2.1052 ± 12.1260	-37.0548	$[-75.1144, -13.0243]$	0.959
$\xi_{180,154}$	-138.6536 ± 57.5605	195.4772	$[-25.0376, 460.2276]$	0.061
$\xi_{154,92}$	42.0072 ± 34.6449	10.7736	$[-22.3436, 56.5905]$	0.774
$\xi_{92,30}$	-81.0404 ± 79.0240	-124.4639	$[-185.5235, -80.1492]$	0.835
$\xi_{30,0}$	377.5806 ± 351.5645	595.2820	$[409.3589, 866.2165]$	0.113

Table A.8: Best-fit results for `base_omegak_CamSpecHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 1.622$, $p = 1.76e - 01$, 1.35σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	211.9556	$[-29.8411, 502.2334]$	0.014
$S_{180,60}$	6694.0882 ± 8822.2915	23719.2671	$[7675.5717, 73031.1322]$	0.131
$S_{180,154}$	2739.9435 ± 3018.8131	3625.1233	$[406.7749, 15623.4577]$	0.453
$S_{154,92}$	2298.0174 ± 3103.8229	7128.0973	$[1943.9653, 24772.1131]$	0.209
$S_{92,30}$	8229.1759 ± 10643.1270	16340.3031	$[6726.9125, 41856.6421]$	0.223
$S_{30,0}$	$39257.1300 \pm 48919.8333$	51705.4424	$[31338.3396, 91825.7121]$	0.295
$\xi_{180,60}$	-2.1052 ± 12.1260	-32.3840	$[-63.4182, -11.6872]$	0.953
$\xi_{180,154}$	-138.6536 ± 57.5605	172.3367	$[-0.9287, 394.2349]$	0.034
$\xi_{154,92}$	42.0072 ± 34.6449	8.3009	$[-20.7864, 42.4205]$	0.835
$\xi_{92,30}$	-81.0404 ± 79.0240	-105.6599	$[-151.8632, -71.0518]$	0.747
$\xi_{30,0}$	377.5806 ± 351.5645	504.0592	$[354.5585, 720.7156]$	0.211

Table A.9: Best-fit results for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

Global tension (ACAT): $T = 3.431$, $p = 9.03e - 02$, 1.69σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	218.3127	$[-17.0392, 508.2853]$	0.006
$S_{180,60}$	6694.0882 ± 8822.2915	24349.6721	$[7615.2117, 72133.8665]$	0.136
$S_{180,154}$	2739.9435 ± 3018.8131	3289.8516	$[501.4208, 16347.5457]$	0.459
$S_{154,92}$	2298.0174 ± 3103.8229	7095.9810	$[2120.8351, 26428.7002]$	0.171
$S_{92,30}$	8229.1759 ± 10643.1270	17049.6488	$[6261.6264, 42388.9107]$	0.224
$S_{30,0}$	$39257.1300 \pm 48919.8333$	54447.2715	$[31086.6010, 96125.6258]$	0.280
$\xi_{180,60}$	-2.1052 ± 12.1260	-33.5933	$[-63.5443, -11.2915]$	0.967
$\xi_{180,154}$	-138.6536 ± 57.5605	172.7077	$[9.0572, 400.9675]$	0.020
$\xi_{154,92}$	42.0072 ± 34.6449	7.2581	$[-21.0722, 40.1496]$	0.853
$\xi_{92,30}$	-81.0404 ± 79.0240	-106.5318	$[-154.6407, -69.6381]$	0.744
$\xi_{30,0}$	377.5806 ± 351.5645	518.2218	$[349.5821, 732.1247]$	0.211

Appendix A.2. Masked sky

In this section the theoretical sampling distributions are constructed with the Planck common mask applied, so the recovered C_ℓ are corrected by f_{sky} before deconvolution.

Appendix A.2.1. MCMC Results

Table A.10: MCMC results for `base_CamSpecHM_TT_low1_lowE` with masked sky.

Global tension (ACAT): $T = 1.965$, $p = 1.50e - 01$, 1.44σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	261.6952	$[-39.0156, 589.9108]$	0.018
$S_{180,60}$	6694.0882 ± 8822.2915	33937.6161	$[10094.0365, 102450.9702]$	0.076
$S_{180,154}$	2739.9435 ± 3018.8131	5200.7872	$[664.4140, 23292.6452]$	0.368
$S_{154,92}$	2298.0174 ± 3103.8229	10088.0224	$[2572.3400, 34735.6347]$	0.138
$S_{92,30}$	8229.1759 ± 10643.1270	23088.7680	$[9244.2759, 60281.6008]$	0.137
$S_{30,0}$	$39257.1300 \pm 48919.8333$	72409.8155	$[41887.8294, 128755.4170]$	0.133
$\xi_{180,60}$	-2.1052 ± 12.1260	-37.1448	$[-77.0816, -13.6870]$	0.962
$\xi_{180,154}$	-138.6536 ± 57.5605	210.0608	$[-7.9407, 473.6089]$	0.047
$\xi_{154,92}$	42.0072 ± 34.6449	8.7805	$[-22.1704, 48.5205]$	0.802
$\xi_{92,30}$	-81.0404 ± 79.0240	-126.3688	$[-182.5317, -81.9472]$	0.844
$\xi_{30,0}$	377.5806 ± 351.5645	603.5009	$[414.7549, 861.6303]$	0.116

Table A.11: MCMC results for `base_plikHM_TT_low1_lowE` with masked sky.

Global tension (ACAT): $T = 1.451$, $p = 1.92e - 01$, 1.30σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	255.8210	$[-34.0357, 574.3206]$	0.019
$S_{180,60}$	6694.0882 ± 8822.2915	24428.7972	$[7160.3979, 75568.3813]$	0.140
$S_{180,154}$	2739.9435 ± 3018.8131	4264.0742	$[600.7927, 18537.1859]$	0.411
$S_{154,92}$	2298.0174 ± 3103.8229	7386.9923	$[1882.9673, 27192.8826]$	0.195
$S_{92,30}$	8229.1759 ± 10643.1270	15784.9025	$[6265.5411, 42504.2238]$	0.254
$S_{30,0}$	$39257.1300 \pm 48919.8333$	57785.5564	$[32425.9781, 102339.8993]$	0.265
$\xi_{180,60}$	-2.1052 ± 12.1260	-31.1565	$[-64.8753, -10.9072]$	0.952
$\xi_{180,154}$	-138.6536 ± 57.5605	193.0745	$[-11.1902, 423.3870]$	0.035
$\xi_{154,92}$	42.0072 ± 34.6449	2.7388	$[-24.4791, 38.4260]$	0.861
$\xi_{92,30}$	-81.0404 ± 79.0240	-104.7700	$[-153.6838, -67.9876]$	0.721
$\xi_{30,0}$	377.5806 ± 351.5645	518.2036	$[341.8661, 753.5033]$	0.237

Table A.12: MCMC results for `base_omegak_CamSpecHM_TT_low1_lowE` with masked sky.

Global tension (ACAT): $T = -1.741$, $p = 8.34e - 01$, 0.21σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	196.3529	$[-33.6480, 499.1034]$	0.013
$S_{180,60}$	6694.0882 ± 8822.2915	16587.5010	$[5049.4661, 50401.6105]$	0.224
$S_{180,154}$	2739.9435 ± 3018.8131	2673.7786	$[349.5165, 13859.4995]$	0.506
$S_{154,92}$	2298.0174 ± 3103.8229	4953.9801	$[1403.6711, 16649.5682]$	0.272
$S_{92,30}$	8229.1759 ± 10643.1270	10932.7937	$[4239.2476, 29132.5902]$	0.392
$S_{30,0}$	$39257.1300 \pm 48919.8333$	39978.8014	$[22623.5916, 74098.2213]$	0.482
$\xi_{180,60}$	-2.1052 ± 12.1260	-25.0097	$[-51.5323, -8.4034]$	0.950
$\xi_{180,154}$	-138.6536 ± 57.5605	145.0064	$[-5.9415, 362.9842]$	0.032
$\xi_{154,92}$	42.0072 ± 34.6449	3.9682	$[-21.0822, 35.0764]$	0.881
$\xi_{92,30}$	-81.0404 ± 79.0240	-87.0262	$[-130.1295, -55.2944]$	0.566
$\xi_{30,0}$	377.5806 ± 351.5645	422.2422	$[281.9245, 629.2318]$	0.388

Table A.13: MCMC results for `base_omegak_plikHM_TTTEEE_lowl_lowE` with masked sky.

Global tension (ACAT): $T = -14.429$, $p = 9.78e - 01$, 0.03σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	199.8608	$[-57.8501, 515.3777]$	0.009
$S_{180,60}$	6694.0882 ± 8822.2915	16593.2759	$[5925.5692, 59966.6153]$	0.184
$S_{180,154}$	2739.9435 ± 3018.8131	2712.8269	$[395.1361, 14323.5511]$	0.500
$S_{154,92}$	2298.0174 ± 3103.8229	4984.4145	$[1606.4512, 20980.0595]$	0.257
$S_{92,30}$	8229.1759 ± 10643.1270	11467.3829	$[4581.3730, 33051.1754]$	0.360
$S_{30,0}$	$39257.1300 \pm 48919.8333$	42300.9829	$[24501.7768, 82246.4234]$	0.432
$\xi_{180,60}$	-2.1052 ± 12.1260	-25.4085	$[-56.4253, -9.2764]$	0.956
$\xi_{180,154}$	-138.6536 ± 57.5605	144.6603	$[-13.2784, 372.6272]$	0.036
$\xi_{154,92}$	42.0072 ± 34.6449	6.5448	$[-21.3855, 37.3271]$	0.868
$\xi_{92,30}$	-81.0404 ± 79.0240	-91.3770	$[-137.0195, -58.4067]$	0.608
$\xi_{30,0}$	377.5806 ± 351.5645	438.1760	$[292.0772, 664.3975]$	0.359

Appendix A.2.2. Best-Fit Results

The mask effect for models with free curvature (Ω_k) is not properly captured since the mask was not designed for those models, while in the MCMC case that effect has been mitigated by the fact that the MCMC ensemble includes a range of cosmological parameters, so the mask effect is averaged over the posterior distribution. In the best-fit case, however, the cosmological parameters are fixed at the best-fit point, so the mask effect is not averaged over the posterior distribution, and therefore it is not properly captured. For that reason the significance of those models with $T \ll 0$ can not be interpreted correctly.

Table A.14: Best-fit results for `base_CamSpecHM_TT_low1_lowE` with masked sky.

Global tension (ACAT): $T = 1.966$, $p = 1.50e - 01$, 1.44σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	260.7976	$[-38.2572, 629.7045]$	0.022
$S_{180,60}$	6694.0882 ± 8822.2915	35725.0265	$[11308.2691, 108785.4640]$	0.073
$S_{180,154}$	2739.9435 ± 3018.8131	5144.2773	$[704.4897, 26362.7403]$	0.358
$S_{154,92}$	2298.0174 ± 3103.8229	10463.2839	$[2893.9033, 39807.8958]$	0.133
$S_{92,30}$	8229.1759 ± 10643.1270	24227.7116	$[9406.5535, 62090.4792]$	0.127
$S_{30,0}$	$39257.1300 \pm 48919.8333$	74487.4419	$[42884.9866, 131879.2429]$	0.114
$\xi_{180,60}$	-2.1052 ± 12.1260	-38.8758	$[-78.4081, -14.6241]$	0.966
$\xi_{180,154}$	-138.6536 ± 57.5605	210.8664	$[7.7966, 508.4919]$	0.051
$\xi_{154,92}$	42.0072 ± 34.6449	7.6378	$[-25.2907, 49.3801]$	0.797
$\xi_{92,30}$	-81.0404 ± 79.0240	-127.2322	$[-184.4908, -84.2408]$	0.862
$\xi_{30,0}$	377.5806 ± 351.5645	611.8525	$[424.7966, 878.9550]$	0.098

Table A.15: Best-fit results for `base_plikHM_TT_low1_lowE` with masked sky.

Global tension (ACAT): $T = 1.281$, $p = 2.11e - 01$, 1.25σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	226.4608	$[-45.2020, 574.4484]$	0.019
$S_{180,60}$	6694.0882 ± 8822.2915	23008.9542	$[7009.3498, 72884.4059]$	0.148
$S_{180,154}$	2739.9435 ± 3018.8131	3582.5904	$[497.1725, 18649.0402]$	0.447
$S_{154,92}$	2298.0174 ± 3103.8229	7215.6867	$[1981.7958, 23609.2984]$	0.184
$S_{92,30}$	8229.1759 ± 10643.1270	15527.5602	$[5771.1736, 39695.6081]$	0.262
$S_{30,0}$	$39257.1300 \pm 48919.8333$	54264.6646	$[31816.0828, 98392.2007]$	0.264
$\xi_{180,60}$	-2.1052 ± 12.1260	-30.0159	$[-60.5086, -10.1636]$	0.951
$\xi_{180,154}$	-138.6536 ± 57.5605	170.6570	$[-6.5249, 426.1746]$	0.042
$\xi_{154,92}$	42.0072 ± 34.6449	4.9295	$[-24.9851, 39.0511]$	0.861
$\xi_{92,30}$	-81.0404 ± 79.0240	-104.5774	$[-152.7222, -67.3813]$	0.720
$\xi_{30,0}$	377.5806 ± 351.5645	500.5559	$[342.1742, 730.5546]$	0.230

Table A.16: Best-fit results for `base_omegak_CamSpecHM_TT_low1_lowE` with masked sky.

Global tension (ACAT): $T = 0.525$, $p = 3.46e - 01$, 0.94σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	199.7885	$[-41.5716, 502.1229]$	0.008
$S_{180,60}$	6694.0882 ± 8822.2915	15186.6432	$[5121.4524, 51624.4976]$	0.227
$S_{180,154}$	2739.9435 ± 3018.8131	2410.4438	$[389.0426, 12858.2779]$	0.528
$S_{154,92}$	2298.0174 ± 3103.8229	4508.5699	$[1226.9281, 17033.6109]$	0.303
$S_{92,30}$	8229.1759 ± 10643.1270	10240.1449	$[4344.4742, 29674.3798]$	0.391
$S_{30,0}$	$39257.1300 \pm 48919.8333$	38598.6800	$[23251.5528, 71396.6366]$	0.514
$\xi_{180,60}$	-2.1052 ± 12.1260	-23.6167	$[-52.2713, -7.9516]$	0.937
$\xi_{180,154}$	-138.6536 ± 57.5605	140.4985	$[-8.6413, 352.1957]$	0.030
$\xi_{154,92}$	42.0072 ± 34.6449	4.8718	$[-18.6700, 34.1368]$	0.892
$\xi_{92,30}$	-81.0404 ± 79.0240	-85.0245	$[-130.9473, -56.0954]$	0.550
$\xi_{30,0}$	377.5806 ± 351.5645	411.5555	$[280.3390, 619.4159]$	0.407

Table A.17: Best-fit results for `base_omegak_plikHM_TTTEEE_lowl_lowE` with masked sky.

Global tension (ACAT): $T = -13.112$, $p = 9.76e - 01$, 0.03σ				
Statistic	Experimental	MCMC Median	68% CI	p -value
C_{180}	-354.1497 ± 171.2834	195.8816	$[-65.4661, 482.5173]$	0.012
$S_{180,60}$	6694.0882 ± 8822.2915	17462.5007	$[5109.3620, 51013.2475]$	0.224
$S_{180,154}$	2739.9435 ± 3018.8131	2706.8102	$[337.4325, 13070.9149]$	0.501
$S_{154,92}$	2298.0174 ± 3103.8229	4905.7765	$[1511.9937, 16861.0841]$	0.259
$S_{92,30}$	8229.1759 ± 10643.1270	11499.9929	$[4374.9723, 29210.2924]$	0.368
$S_{30,0}$	$39257.1300 \pm 48919.8333$	41696.6899	$[24022.0627, 75140.3093]$	0.455
$\xi_{180,60}$	-2.1052 ± 12.1260	-25.5638	$[-52.9067, -8.0940]$	0.943
$\xi_{180,154}$	-138.6536 ± 57.5605	140.1252	$[-14.3516, 359.1568]$	0.034
$\xi_{154,92}$	42.0072 ± 34.6449	4.2472	$[-19.6041, 35.6554]$	0.883
$\xi_{92,30}$	-81.0404 ± 79.0240	-89.4519	$[-131.2958, -57.3051]$	0.591
$\xi_{30,0}$	377.5806 ± 351.5645	437.4789	$[283.1412, 631.8542]$	0.367